Freelancer Marketplace Behaviour Analysis

Introduction

This project analyzes the behavior of users in a freelancer marketplace platform. It focuses on how clients post projects, how freelancers bid, and how contracts and reviews are generated. The goal is to understand patterns in hiring, pricing, and freelancer performance using SQL-based data analysis.

Objective

- To analyze freelancer and client interactions
- To study bidding patterns and project trends
- To evaluate freelancer performance using ratings
- To identify factors affecting project completion and success

Tools Used

- MySQL Workbench
- SQL (Structured Query Language)
- Microsoft Excel (for dataset creation and preprocessing)

Database Design

1. Users

- Purpose: Stores all users (clients & freelancers)
- Fields: user_id (PK), name, email, country, user_type, join_date

2. Freelancer_Profiles

- Purpose: Freelancer-specific info
- Fields: profile_id (PK), freelancer_id (FK → Users), skills, experience_level, hourly_rate

3. Projects

- Purpose: Client projects
- Fields: project_id (PK), client_id (FK → Users), title, description, budget, status, post_date, deadline

4. Bids

- Purpose: Freelancer proposals for projects
- Fields: bid_id (PK), project_id (FK → Projects), freelancer_id (FK → Users), bid_amount, proposal_text, bid_date

5. Contracts

- Purpose: Accepted bid agreement
- Fields: contract_id (PK), project_id (FK → Projects), freelancer_id (FK → Users), client_id (FK → Users), start_date, end_date, payment_status, contract_amount

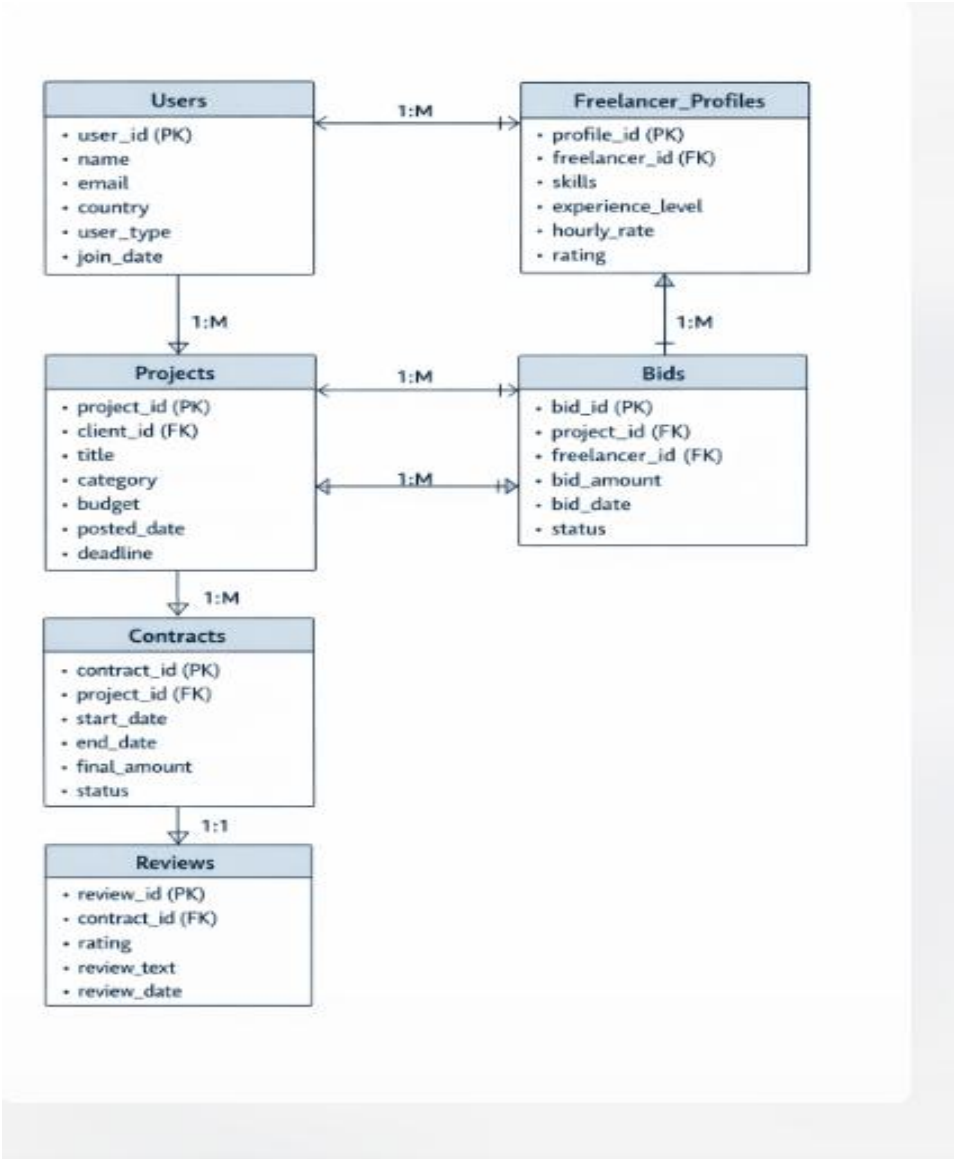
6. ***Reviews***

- Purpose: Client feedback for freelancer
- Fields: review_id (PK), contract_id (FK → Contracts), rating, comments, review_date

Relationships:

- Users → Freelancer_Profiles (1:1)
- Users → Projects (1:M)
- Projects → Bids (1:M)
- Projects + Users → Contracts (1:1)
- Contracts → Reviews (1:1)

Relational Diagram



Dataset Description

The dataset contains information about users, freelancers, projects, bids, contracts, and reviews on a freelancing platform. It tracks user details, project postings, freelancer proposals, contract agreements, and client feedback. It can be used to analyze skills, budgets, bidding trends, and project outcomes. Additionally, it provides insights into freelancer performance, client satisfaction, and project completion timelines. The dataset can also help identify popular project categories, average project durations, and pricing patterns across different industries. It is useful for studying marketplace dynamics, evaluating talent demand and supply, and making data-driven decisions for improving platform efficiency.

SQL Queries with Questions

List All Freelancers

1**) Which users are registered as freelancers on the platform?**
Query:

```
SELECT * FROM Users

WHERE user_type = 'Freelancer';
```

Insight: Identifies all freelancers to analyze their skills, availability, and profiles.

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

user_id	name	email	country	user_type	join_date
6	Ali Hassan	ali@gmail.com	UAE	freelancer	2045-06-06
7	Neha Singh	neha@gmail.com	India	freelancer	2045-09-05
10	Emily Clark	emily@gmail.com	Canada	freelancer	2045-00-07
NULL	NULL	NULL	NULL	NULL	NULL

List All Clients

2)Which users are registered as clients on the platform?
Query:

```
SELECT * FROM Users
WHERE user_type = 'Client';
```

Insight: Shows all clients, useful to track project postings and engagement.

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

user_id	name	email	country	user_type	join_date
5	David Lee	david@gmail.com	UK	client	2045-02-08
8	Robert Brown	robert@gmail.com	USA	client	2045-10-08
NULL	NULL	NULL	NULL	NULL	NULL

Count of Projects Per Client

3) How many projects has each client posted?
Query:

```
SELECT client_id, COUNT(*) AS total_projects
FROM Projects
GROUP BY client_id;
```

Insight: Identifies the most active clients on the platform.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

client_id	total_projects
5	2
8	2

Highest Bid for Each Project

4) What is the highest bid for each project?
Query:

```
SELECT project_id, MAX(bid_amount) AS highest_bid
FROM Bids
GROUP BY project_id;
```

Insight: Shows budget competitiveness and top bids per project.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

project_id	highest_bid
103	280.00
104	140.00
108	330.00
109	110.00

Average Hourly Rate of Freelancers

5)What is the average hourly rate of freelancers?
Query:

SELECT AVG(hourly_rate) AS avg_hourly_rate
FROM Freelancer_Profiles;

Insight: Useful for clients to benchmark their budgets and pricing expectations.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg_hourly_rate
21.500000

Average Rating per Freelancer

6)Find average rating of each freelancer based on reviews

Query:

```
SELECT c.freelancer_id, AVG(r.rating) AS avg_rating
FROM Reviews r
JOIN Contracts c ON r.contract_id = c.contract_id
GROUP BY c.freelancer_id;
```

Insight:
Provides feedback-based performance evaluation for freelancers.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

freelancer_id	avg_rating
6	5.00000
7	4.00000

Projects with Average Bid Amount

7) What is the average bid amount for each project?
Query:

```
SELECT project_id, AVG(bid_amount) AS avg_bid
FROM Bids
GROUP BY project_id;
```

Insight: Shows general pricing trends and helps clients set competitive budgets.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

project_id	avg_bid
103	280.000000
104	140.000000
108	330.000000
109	110.000000

Freelancers Who Never Received a Contract

8)Which freelancers have not been awarded any contract yet?
Query:

```
SELECT u.user_id, u.name
FROM Users u
LEFT JOIN Contracts c ON u.user_id = c.freelancer_id
WHERE u.user_type = 'Freelancer' AND c.contract_id IS NULL;
```

Insight: Identifies underutilized freelancers for better engagement strategies.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

user_id	name
10	Emily Clark

Top Clients by Number of Projects

9)Which clients have posted the most projects?
Query:

```
SELECT client_id, COUNT(*) AS total_projects
FROM Projects
GROUP BY client_id
ORDER BY total_projects DESC
LIMIT 5;
```

Insight: Identifies the most active clients on the platform, which can help target engagement and marketing strategies.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

client_id	total_projects
5	2
8	2

Average Bid Amount per Freelancer

10)What is the average bid amount submitted by each freelancer?
Query:

```
SELECT freelancer_id, AVG(bid_amount) AS avg_bid
FROM Bids
GROUP BY freelancer_id;
```

Insight: Provides insight into typical freelancer pricing and bidding patterns, helping clients evaluate fair bids.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

freelancer_id	avg_bid
6	305.000000
7	125.000000

Top-Rated Freelancers

11)Which freelancers have the highest average rating?
Query:

```
SELECT c.freelancer_id, AVG(r.rating) AS avg_rating
FROM Reviews r
JOIN Contracts c ON r.contract_id = c.contract_id
GROUP BY c.freelancer_id
ORDER BY avg_rating DESC
LIMIT 5;
```

Insight: Highlights the most reliable and high-performing freelancers, aiding clients in selection.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

freelancer_id	avg_rating
6	5.00000
7	4.00000

Freelancer Distribution by Experience Level

12)How many freelancers belong to each experience level?
Query:

```
SELECT experience_level, COUNT(*) AS total_freelancers
FROM Freelancer_Profiles
GROUP BY experience_level;
```

Insight: Helps understand the distribution of beginners, intermediates, and experts on the platform for project allocation.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

experience_level	total_freelancers
Expert	2
Beginner	2
Intermediate	2

Freelancer Activity in Last 30 Days

13)How many bids has each freelancer submitted in the last 30 days?
Query:

```
SELECT freelancer_id, COUNT(*) AS bids_last_month
FROM Bids
WHERE bid_date >= DATE_SUB(CURDATE(), INTERVAL 30 DAY)
GROUP BY freelancer_id;
```

Insight: Identifies active freelancers and measures recent engagement on the platform.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

freelancer_id	bids_last_month
6	2
7	2

Freelancers with Most Skills Listed

14)Which freelancers have listed the highest number of skills?
Query:

```
SELECT freelancer_id, LENGTH(skills) - LENGTH(REPLACE(skills, ',', '')) + 1 AS skill_count
FROM Freelancer_Profiles
ORDER BY skill_count DESC
LIMIT 5;
```

Insight: Highlights versatile freelancers with multiple skills, useful for multi-domain project assignments.

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

freelancer_id	skill_count
6	1
7	1
10	1
6	1
7	1

Key Insights

- Most active freelancers have diverse skills and moderate hourly rates.
- Projects with clear budgets and deadlines attract more bids.
- Higher-rated freelancers win contracts more frequently.

Conclusion

The platform enables efficient matching between clients and freelancers. Analyzing bids, contracts, and reviews helps understand trends, performance, and user behavior for better decision-making.

Future Scope

- Implement AI-based project–freelancer recommendation system.
- Predict project success and freelancer performance using analytics.
- Introduce dynamic pricing and skill-based matchmaking.
- Expand reporting dashboards for clients and freelancers.